

Evaluation Report

System: RA Electronic Communications Law RAG assistant

Providers compared: OpenAI `gpt-4o-mini`, Google Gemini `gemini-3.6-flash`, Groq `qwen/qwen3.8-27b` (xAI Grok `grok-4.3` configured but invalid API key — excluded from this run)

Shared pipeline: article-aware chunking · OpenAI `text-embedding-3-large` (1536) · Neon pgvector · grounded prompt · confidence gate · citation filter · Groq `context/ max_tokens budget for free-tier TPM`

Gold set: `eval/questions.json` (19 claim-based items: 6 HY + 7 EN answerable incl. synthesis; 6 adversarial oos / near-miss / underspecified). Atomic `gold_key_points + must_not`; self-verified against `data/law.md` (not bar-certified). **Scoring:** fixed `gpt-4o-mini` LLM judge on top of the heuristic claim match + `must_not` checks (answer accuracy and hallucination rate are judge-refined; citation accuracy and Recall@k are deterministic)

1. Benchmark results (per model)

Metric	OpenAI gpt-4o-mini	Gemini 3.6 Flash	Groq Qwen3.8-27B	Grok grok-4.3
Answer accuracy (mean)	96.1%	95.8%*	89.5%	— (invalid key)
Citation accuracy (mean)	94.7%	70.0%*	61.8%	—
Hallucination rate (mean)	5.0%	6.7%*	9.6%	—
Retrieval Recall@k (shared)	94.7%†	94.7%†	94.7%†	—
TTFT p50 / p95 (ms)	797 / 2711	4373 / 10764	29985 / 33759	—
Total latency p50 / p95 (ms)	1735 / 4265	4933 / 11086	30326 / 35532	—
Avg prompt + completion tokens	3299 + 94	5491 + 176	5085 + 320	—
Est. cost @ paid rates (USD, full run)	\$0.0105	\$0.0717	\$0.1016	—
Failure rate (rate limit / timeout / malformed)	0%	21.1%	0%	—

* Gemini accuracy / citation / hallucination averages are over **successful** completions only; 4/19 calls failed (all `503 Service Unavailable` from the Gemini API, recorded as `error: en-04, en-07, hy-02, hy-03`).

† Shared retrieval; adversarial items with empty `gold_articles` count as perfect recall. On **answerable** items only, Recall@k ≈ **92.3%** (12/13 — only hy-02 misses Art. 17).

Run id: `cmuf7bi4d0000jdyrtdc8yc4`
Date: 2026-09-24
Raw export: [docs/benchmark/run-cmuf7bi4d0000jdyrtdc8yc4.json](#)

Slice highlights

Slice	OpenAI ans / cite / hall	Gemini ans / cite / hall (fail)	Groq ans / cite / hall
Armenian answerable (6)	87.5% / 83.3% / 12.5%	100% / 100% / 0% (33% fail)	75.0% / 70.8% / 12.5%
English answerable (7)	100% / 100% / 2.9%	87.5% / 90.0% / 20.0% (29% fail)	92.9% / 78.6% / 15.4%
Adversarial (6)	100% / 100% / 0%	100% / 33.3% / 0% (0% fail)	100% / 33.3% / 0%

All three models refuse / clarify correctly on every adversarial item. The difference is citation discipline: Gemini and Groq still attach retrieved article numbers to their refusals on adv-02...05 (gold expects none), so their adversarial citation accuracy drops to 33%; OpenAI cites nothing on those items. Gemini's perfect Armenian slice is over only 4 completed items — both failed HY calls (hy-02, hy-03) include the hardest retrieval case. The hardest items for everyone are hy-02 (retrieval miss on Art. 17) and hy-06 (Art. 2 definitions retrieved last, behind Arts 59 / 31 / 53).

How metrics are computed

Metric	Method
Answer accuracy	Claim coverage vs atomic <code>gold_key_points</code> , refined by the <code>gpt-4o-mini</code> judge; adversarial = refuse / clarify by subtype

Metric	Method
Citation accuracy	Precision of cited article numbers vs <code>gold_articles</code> (adversarial: empty citations score 1)
Hallucination rate	Citations outside retrieved set and/or <code>must_not</code> trap hits, refined by the judge for unsupported claims in prose
Latency	Streaming TTFT + total wall time per completion
Cost	Tokens × public paid list prices in <code>src/lib/llm/pricing.ts</code>
Failure rate	<code>status ≠ ok</code> (<code>rate_limit</code> / <code>timeout</code> / <code>malformed</code> / <code>error</code>)

2. Recommendation

Recommended default for production standardization: OpenAI `gpt-4o-mini`.

Reasons (tied to this run):

- Best reliability** — 0% failure across all 19 questions; Gemini lost 21% of calls to upstream 503s.
- Best citation discipline** — 94.7% citation accuracy overall and 100% on adversarial items (no article cites on refusals), versus 70% (Gemini) and 62% (Groq). This matters most for a legal desk tool.
- Best latency / cost** — TTFT p50 ~0.8s, total p50 ~1.7s; ~\$0.01 for the full 19×1 eval at paid rates vs ~\$0.07–\$0.10 for Gemini/Groq.
- Top answer accuracy** — 96.1% overall, level with Gemini's successful-call average (95.8%) and ahead of Groq (89.5%), with the lowest hallucination rate (5.0%).

When to prefer another model:

- Gemini Flash** — matches OpenAI on answer accuracy when it completes and was perfect on the Armenian items it finished; not yet reliable enough (21% failures) for a full desk load, and it cites articles on refusals.
- Groq Qwen** — no failures, but ~30s p50 latency and the weakest citation / hallucination numbers; only suitable for offline batch work with citation filtering on.
- Grok** — re-test after installing a valid `GR0K_API_KEY` from [console.x.ai](#) (do not reuse a Groq `gsk_` key).

Caveats:

- The judge is `gpt-4o-mini`, the same model as the recommended provider, so answer accuracy / hallucination may carry some preference toward OpenAI-style answers. Citation accuracy, Recall@k, latency, cost and failure rate are deterministic and not affected; OpenAI also led on answer accuracy in the previous heuristic-only run.
- Retrieval is no longer the main cap (answerable Recall@k 92.3%); remaining errors are mostly citation precision and ranking (hy-06).
- Groq free-tier TPM (~8k) requires capped `max_tokens` and truncated context (`src/lib/llm/groq.ts` , `src/lib/rag/generate.ts`).

3. Trade-offs

Latency vs accuracy

- OpenAI wins the interactive Ask path (sub-2s p50, ~4s p95) and is also the most accurate in this run.
- Groq is the slowest by far (~30s p50 TTFT, thinking-style completions) without an accuracy advantage.
- Gemini sits in the middle on latency (~5s p50) when healthy, but upstream 503s dominate its failure rate.

Armenian language handling

- Corpus is Armenian-only; EN questions use multilingual embeddings + **always-on EN→HY query translation** with a statutory glossary (`src/lib/rag/retrieve.ts`).
- Generation language follows the question (`src/lib/rag/language.ts`).
- In this run the **Armenian** slice is now the harder one: EN answerable retrieval is 100% (7/7), HY answerable is 83% (5/6). OpenAI scores 87.5% on HY vs 100% on EN; Groq 75% vs 93%.

Rate-limit reliability

- Generation, embeddings, and the optional judge use exponential backoff (`src/lib/util/backoff.ts`).
- Gemini: non-streaming fallback after intermittent stream parse failures (`src/lib/llm/gemini.ts`). This run's failures were upstream `503` errors, not free-tier 429s.

- Groq: `max_tokens=768` + ~8k-char context budget so free-tier TPM does not 413 on declared ceilings.
- Failures are recorded per result (not silently dropped).

4. Known limitations

- Single law / single jurisdiction; no multi-document routing.
- PDF/ARLIS extract may miss amendment nuance if `law.md` is stale vs latest incorporation.
- Parent chunk embeddings truncate very long articles (~2500 chars of body); children mitigate but parent-only hits can be incomplete.
- Distance-based confidence gate helps only for *weak* neighbors; near-topic adversarials still retrieve related articles — refusal relies on the generator, and some generators still cite those articles.
- Soft grounding: citation **cards** are filtered to the retrieved set; answer prose is prompt-only.
- Provider availability (Gemini 503s) and invalid keys (Grok) can dominate failure rate unrelated to model quality.
- Judge and recommended provider are the same model family (see caveats in §2).
- No human legal review loop; outputs are assistive, not advice.

5. What we would do with more time

- Install a valid xAI Grok key and complete the fourth column.
- Use a judge from a different provider (or a panel) to remove self-preference risk.
- Try **BGE-M3** or Cohere multilingual embeddings; fix the remaining HY retrieval miss (hy-02 → Art. 17) and ranking of definitions (hy-06 → Art. 2).
- Stronger hybrid retrieval (BM25 / Postgres FTS on Armenian stems + vectors).
- Structured citation JSON schema / constrained decoding per provider; drop citations on refusals.
- Lawyer review of the claim-based gold set; separate HY-only and EN-only leaderboards.
- Persistent caching of embeddings and retrieval for Ask UI.
- UI: article deep-links into `law.md` offsets; export signed PDF memos.

Appendix — gold set coverage (claim-based, article-verified)

Slice	Count	IDs / articles
Armenian answerable	6	hy-01...06 → Arts 30, 17, 33, 22–23, 45, 2
English answerable	7	en-01...07 → Arts 3, 26, 13, 48, 39, 40, 27
Adversarial	6	adv-01...03 oos · adv-04...05 near_miss · adv-06 underspecified
Total	19	

Claims and citations checked against `data/law.md` (ARLIS extract). Not bar-certified.

Ops notes (this run)

1. Gemini / Groq / OpenAI keys were set; Grok key still invalid.
2. Benchmark tab → OpenAI + Gemini + Groq → Run (judge) on the 19-item claim-based gold set produced run `cmuf7bi4d000jdtyrtdc8yc4` (2026-09-24, ~13 min).
3. Previous run `cmucogmp4000daym3sge3h36i` (20 items, old gold wording, heuristic only) is kept in `docs/benchmark/` for reference; its numbers are not comparable to this run.